

ECHS FRAUD ANALYTICS

Geo-Spatial Fraud Clustering Analysis

MODULE 9 REPORT — FY 2021-2026 EDITION (LAST 5 YEARS)

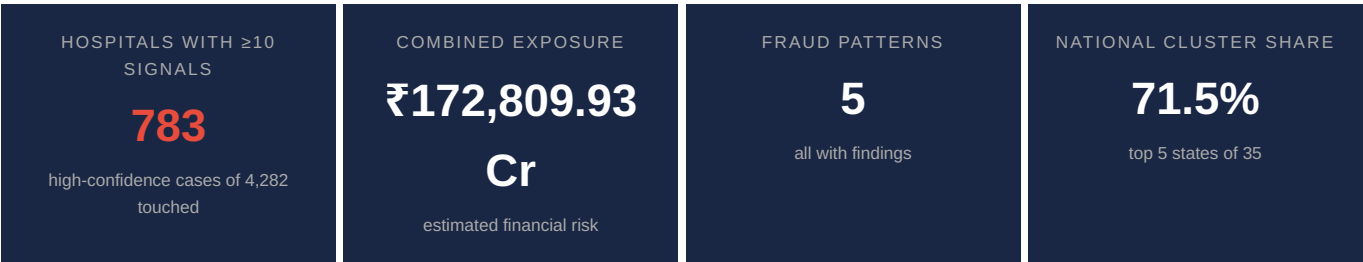
CLASSIFICATION	PERIOD	HOSPITALS SCANNED	PATTERNS RUN	PEAK DEDUCTION
RESTRICTED	FY 2021-26	4,282	5 Patterns	19.23%

IIT KANPUR — Data Analytics & Fraud Intelligence Division

1 July 2026 | Ex-Servicemen Contributory Health Scheme (ECHS)

EXECUTIVE SUMMARY

This report analyses the **geographic concentration of ECHS fraud across India**, organised by ECHS's own regional/state geography: which regions and states are fraud hotspots, how concentrated hospital chains and referral relationships are, and where flagged claim activity clusters nationally. Pattern 1 establishes the region-level deduction baseline (settlement_stat, last 5 financial years, FY 2021-22 to FY 2025-26). Patterns 2–5 build on 20 named fraud-detection signals already established elsewhere in the ECHS fraud-analytics programme, re-mapping them geographically without running any new fraud-detection queries. The headline finding: ECHS fraud is not spread evenly across the country — **Haryana, Punjab, Uttar Pradesh, Rajasthan, Delhi account for 71.5% of all cross-pattern flagged exposure nationally** (Pattern 3).



FIVE GEO-SPATIAL FRAUD PATTERNS — SUMMARY

#	Pattern	What It Detects	Flagged / Analysed
1	Regional Fraud Hotspots	Deduction rate and total claims by ECHS region (settlement baseline)	31 regions — highest 19.23% deduction
2	Cross-Pattern Hospital Concentration	20 named fraud-detection signals — hospitals triggering ≥10 signals are high-priority	783 hospitals ≥10 signals; 4,282 touched by ≥1
3	Cross-Pattern Regional & State Clustering	Geographic rollup of flagged cases vs. total hospitals and claims	31 regions, 35 states — top 5 = 71.5% of exposure
4	Cross-Pattern Hospital Chain Detection	Multi-unit hospital brands — total vs. flagged units per chain	564 chains — 9/10 flagged in top chain
5	Cross-Region Referral Usage	Single-hospital referral funnelling — Tukey-fence outliers by exposure	10 statistical outliers of 78 pairs

IMMEDIATE RECOMMENDED ACTIONS

- Joint audit of hospitals triggering ≥10 of the 20 named fraud signals** (783 hospitals) — these are corroborated by multiple independent statistical methods, not a single pattern's threshold; prioritise above single-signal outliers.
- Field audit deployment to R-6 (Lucknow) and R-1 (Delhi)** — the two regions topping the baseline deduction-rate and cross-pattern exposure rankings respectively.
- Corporate-group-level audit of multi-unit chains**, starting with METRO HEART (9 of 10 units already flagged) — audit the legal-entity group as a whole, not branch-by-branch.
- Recovery review for state-level concentration** — Haryana accounts for the largest share of combined cross-pattern flagged exposure (₹45,394.40 Cr) and should anchor the next regional audit cycle.
- Audit the 10 referral pairs exceeding the Tukey-fence referral-count threshold** (47,851 referrals; Table 5.1), starting with VidyaDhar Nagar (VidyaDhar Nagar) → MANIPAL HEALTH ENTERPRISES PRIVATE LIMITED (JAIPUR) — a single referral source sending the overwhelming majority of its patients to one out-of-area hospital is a classic referral-syndicate signature.
- Extend geo-mapping to the remaining fraud-detection analyses** once their exports carry a numeric office hospital identifier — a handful of analyses (identity/duplicate-claims, length-of-stay, diagnostic overutilisation, network collusion) cannot yet be geo-mapped because their existing outputs either have no hospital-level rollup or use free-text identifiers incompatible with the office master's numeric ID space, the same ID-mismatch issue this report's predecessor flagged.

PATTERN 1

Regional Fraud Hotspot Analysis
settlement_stat — all 31 ECHS regions, last
5 years

REGIONAL FRAUD HOTSPOTS

Description: Aggregates claims in settlement_stat by ECHS region code (SS_REGION_ID) — last 5 financial years (FY 2021-22 to FY 2025-26). Shows the financial baseline for each ECHS region: total settlement volume, claimed and approved amounts, and the deduction rate (Approved ÷ Claimed). Regions with high deduction rates signal systematic over-billing in that geography. SS_REGION_ID is the CGHS Region/City code space, annotated with the matching CGHS region name below.

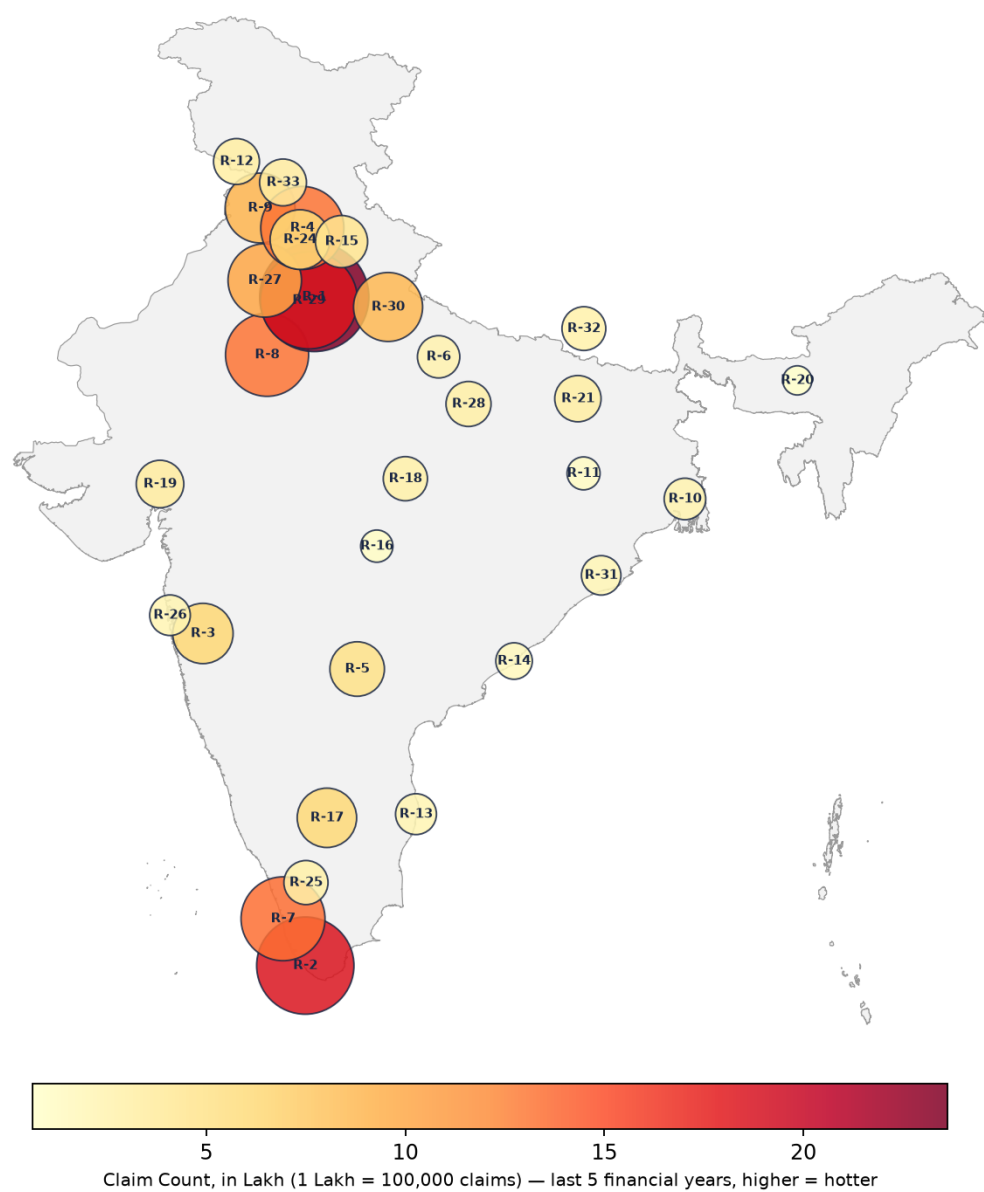
Table 1.1 — All 31 ECHS Regions, Sorted by Deduction Rate

Region	Hospitals	Total Claims	Claimed	Approved	Ded %
R-6 (Lucknow)	110	259,235	₹783.03 Cr	₹632.47 Cr	19.23%
R-13 (Chennai)	76	233,901	₹368.62 Cr	₹303.21 Cr	17.75%
R-8 (Jaipur)	261	1,321,549	₹3192.92 Cr	₹2647.36 Cr	17.09%
R-25 (Coimbatore)	81	290,662	₹373.56 Cr	₹318.53 Cr	14.73%
R-3 (Pune)	211	644,364	₹984.26 Cr	₹847.63 Cr	13.88%
R-20 (Guwahati)	65	63,418	₹114.24 Cr	₹99.23 Cr	13.14%
R-5 (Secunderabad)	129	504,364	₹676.12 Cr	₹593.67 Cr	12.19%
R-16 (Nagpur)	70	102,454	₹157.87 Cr	₹139.42 Cr	11.69%
R-26 (Mumbai)	63	229,166	₹247.57 Cr	₹218.88 Cr	11.59%
R-29 (Delhi 2)	325	1,847,943	₹2942.95 Cr	₹2603.02 Cr	11.55%
R-18 (Jabalpur)	138	292,164	₹362.05 Cr	₹320.37 Cr	11.51%
R-27 (Hisar)	208	1,007,338	₹2677.03 Cr	₹2381.22 Cr	11.05%
R-11 (Ranchi)	58	112,520	₹129.76 Cr	₹115.88 Cr	10.69%
R-17 (Bangalore)	139	620,357	₹816.45 Cr	₹729.70 Cr	10.63%
R-10 (Kolkata)	76	245,419	₹552.73 Cr	₹497.33 Cr	10.02%
R-32 (Nepal)	30	285,322	₹363.84 Cr	₹328.19 Cr	9.80%
R-24 (Ambala)	196	627,355	₹1910.18 Cr	₹1732.91 Cr	9.28%
R-30 (Bareilly)	157	876,537	₹1309.81 Cr	₹1188.77 Cr	9.24%
R-1 (Delhi)	392	2,362,788	₹5034.50 Cr	₹4576.22 Cr	9.10%
R-14 (Visakhapatnam)	51	163,899	₹182.40 Cr	₹165.85 Cr	9.08%
R-31 (Bhubaneswar)	34	210,025	₹173.22 Cr	₹158.49 Cr	8.51%
R-19 (Ahmedabad)	121	356,456	₹467.64 Cr	₹428.76 Cr	8.31%
R-15 (Dehradun)	84	452,411	₹663.19 Cr	₹609.77 Cr	8.05%
R-21 (Patna)	71	332,227	₹413.82 Cr	₹380.68 Cr	8.01%
R-2 (Trivandrum)	104	1,854,159	₹1149.20 Cr	₹1065.70 Cr	7.27%

Region	Hospitals	Total Claims	Claimed	Approved	Ded %
R-28 (Allahabad)	79	308,037	₹573.43 Cr	₹532.27 Cr	7.18%
R-33 (Yol)	51	343,079	₹305.56 Cr	₹283.95 Cr	7.07%
R-7 (Kochi)	123	1,351,496	₹1202.53 Cr	₹1119.22 Cr	6.93%
R-9 (Jalandhar)	229	900,746	₹3990.77 Cr	₹3725.56 Cr	6.65%
R-4 (Chandimandir)	160	1,321,761	₹3496.72 Cr	₹3294.65 Cr	5.78%
R-12 (Jammu)	108	324,232	₹645.45 Cr	₹610.13 Cr	5.47%

Figure 1.1 — Regional Claim-Volume Heatmap, All India (Last 5 FYs)

ECHS Claim Volume by CGHS Region — All India



Each bubble is one **CGHS Region** — the city-based location/rate-region classification ECHS's own claims data uses internally (e.g. R-9 = the Jalandhar region; see the CGHS Region Master for the full list). ECHS itself is headquartered in New Delhi and administers all 31 regions nationally; CGHS naming is simply the geographic key the underlying claims table uses. Bubble size and colour both scale with claim count, in Lakh — darker/larger = higher claim volume passing through that region. The map shows India's full national boundary, including Jammu & Kashmir, Ladakh and Aksai Chin, per India's official territorial extent.

Key Findings

- R-6 (Lucknow)** has the highest deduction rate at **19.23%** — ₹783.03 Cr claimed vs. ₹632.47 Cr approved across 110 hospitals and 259,235 total claims.
- R-8 (Jaipur)** has the largest absolute deduction — ₹545.56 Cr deducted from ₹3192.92 Cr claimed (17.09% rate) across 261 hospitals.

PATTERN 2

CROSS-PATTERN HOSPITAL CONCENTRATION

Independent-Method Corroboration
Hospital-level fraud signals, geo-mapped

Description: Each row is a hospital showing its **Total Claims** (real settlement volume, last 5 FYs), how many of the 20 named fraud-detection signals it triggered, and its **Combined Exposure** (total flagged amount in ₹ Cr). The signals used are: Revenue Inflation Outliers, Procedure Upcoding, Doctor Referral Concentration, Beneficiary Utilisation Outliers, Repeat Admission Outliers, Procedure Mislabelling, Emergency Admission Misuse, Package/Item Billing Anomaly, Time-Series Billing Surge, Hospital Specialty Misuse, High-Value Claim Risk Scoring, Pre-Authorization Deviation, Claim Processing / Overbilling Pattern, Policy Abuse Pattern, Implausible Claim Volume, Extreme Deduction Outliers, Same-Day Zero-Stay High-Value IPD, Hospital Escalation (YoY Ratio Rise), Room Entitlement Mismatch, Emergency Claim Flag Abuse. A hospital flagged by many unrelated signals is a far stronger fraud indicator than a high score on any single one, because each signal tests a different part of the claims lifecycle. Table 2.2 names exactly which signals fired for each top hospital.

Table 2.1 — Top 20 Hospitals by Number of Distinct Signals Triggered

ID	Hospital	State	Signals Triggered	Total Claims	Combined Exposure
3019	SAKRA WORLD HOSPITAL	Karnataka	18 / 20	9,199	₹89.89 Cr
3149	VIJAY HOSPITAL	Haryana	16 / 20	21,815	₹3,226.66 Cr
2856	VIRAT HOSPITAL	Haryana	16 / 20	20,383	₹973.42 Cr
2826	HUMAN CARE MEDICAL CHARITABLE TRUST	Delhi	16 / 20	84,776	₹807.22 Cr
1197	MOOLCHAND HOSPITAL - Lajpat Nagar -	Delhi	16 / 20	10,232	₹72.19 Cr
399	MANIPAL HEALTH ENTERPRISES PRIVATE L	Rajasthan	15 / 20	167,341	₹2,117.16 Cr
183	MEDANTA THE MEDICITY - GURGAON	Haryana	15 / 20	229,963	₹2,039.82 Cr
274	METRO HEART INSTITUTE - FARIDABAD	Haryana	15 / 20	39,091	₹1,649.62 Cr
514	PARK HOSPITAL (A UNIT OF UMKAL HEALT	Haryana	15 / 20	33,098	₹1,599.85 Cr
2766	THE SIGNATURE HOSPITAL (A UNIT OF PA	Haryana	15 / 20	22,800	₹1,180.80 Cr
2694	STAR HOSPITAL (A UNIT OF OM MEDICENT	Rajasthan	15 / 20	4,759	₹1,151.88 Cr
63	KALRA HOSPITAL SRCNC PVT LTD - KIRTI	Delhi	15 / 20	8,235	₹721.41 Cr
21	PARAS HEALTHCARE PVT LTD - SUSHANT L	Haryana	15 / 20	83,065	₹579.79 Cr
1585	YATHARTH WELLNESS SUPER SPECIALITY H	Uttar Pradesh	15 / 20	14,070	₹451.95 Cr
1535	ASTER DM HEALTHCARE LIMITED	Kerala	15 / 20	31,311	₹325.76 Cr
28	S K HOSPITAL	Kerala	15 / 20	125,652	₹276.00 Cr
2163	MANIPAL HOSPITALS PRIVATE LIMITED	Haryana	15 / 20	101,777	₹248.83 Cr
1676	METRO HOSPITAL & HEART INSTITUTE - H	Uttarakhand	15 / 20	3,402	₹211.22 Cr
111	Care Hospital (Quality Care India Lt	Andhra Pradesh	15 / 20	28,397	₹198.96 Cr
3058	MAZUMDAR SHAW MEDICAL CENTER (A UNIT	Karnataka	15 / 20	10,666	₹117.66 Cr

Table 2.2 — Which Signals Fired (Top 10 Hospitals)

Hospital	Signals Triggered (Named)
SAKRA WORLD HOSPITAL	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Package/Item Billing Anomaly, Policy Abuse Pattern, Pre-Authorization Deviation, Procedure Mislabelling, Procedure Upcoding, Revenue Inflation Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD, Time-Series Billing Surge
VIJAY HOSPITAL	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Specialty Misuse, Implausible Claim Volume, Package/Item Billing Anomaly, Policy Abuse Pattern, Procedure Mislabelling, Procedure Upcoding, Revenue Inflation Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD, Time-Series Billing Surge
VIRAT HOSPITAL	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Specialty Misuse, Implausible Claim Volume, Package/Item Billing Anomaly, Policy Abuse Pattern, Procedure Mislabelling, Procedure Upcoding, Revenue Inflation Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD, Time-Series Billing Surge
HUMAN CARE MEDICAL CHARITABLE	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Policy Abuse Pattern, Pre-Authorization Deviation, Procedure Mislabelling, Repeat Admission Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD, Time-Series Billing Surge
MOOLCHAND HOSPITAL - Lajpat Na	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Specialty Misuse, Implausible Claim Volume, Package/Item Billing Anomaly, Policy Abuse Pattern, Procedure Mislabelling, Procedure Upcoding, Revenue Inflation Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD, Time-Series Billing Surge
MANIPAL HEALTH ENTERPRISES PRI	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Policy Abuse Pattern, Pre-Authorization Deviation, Procedure Mislabelling, Repeat Admission Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD
MEDANTA THE MEDICITY - GURGAON	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Policy Abuse Pattern, Pre-Authorization Deviation, Procedure Mislabelling, Repeat Admission Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD
METRO HEART INSTITUTE - FARIDA	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Policy Abuse Pattern, Pre-Authorization Deviation, Procedure Mislabelling, Repeat Admission Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD
PARK HOSPITAL (A UNIT OF UMKAL	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Policy Abuse Pattern, Pre-Authorization Deviation, Repeat Admission Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD, Time-Series Billing Surge
THE SIGNATURE HOSPITAL (A UNIT	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Policy Abuse Pattern, Pre-Authorization Deviation, Procedure Mislabelling, Repeat Admission Outliers, Room Entitlement Mismatch, Time-Series Billing Surge

Key Findings

SAKRA WORLD HOSPITAL (BANGALORE, Karnataka) is flagged by **18 of the 20 named signals** (see Table 2.2) — 9,199 total claims handled, combined flagged exposure of ₹89.89 Cr.

Hospitals triggering 10+ of the 20 named signals should be prioritised for joint audit ahead of hospitals with a high score on any single signal alone — the convergence of unrelated statistical tests on the same facility sharply reduces the chance of a false positive.

PATTERN 3

Geographic Rollup of All Flagged Cases

By ECHS region & state

CROSS-PATTERN REGIONAL & STATE CLUSTERING

Description: Rolls up the cross-pattern hospital index by ECHS region and by state, to show where flagged hospitals concentrate geographically. **Total Hospitals** is every empanelled hospital in that region/state; **Flagged Hospitals** is the subset triggering at least one of the 20 named signals. **Distinct Signals** is how many of the 20 independent methods flagged activity in that region/state. **Combined Exposure** is the total flagged amount in ₹ Cr.

Table 3.1 — ECHS Regions Ranked by Combined Cross-Pattern Exposure

Region	Total Hospitals	Flagged Hospitals	Distinct Signals	Total Claims	Combined Exposure
R-1 (Delhi)	429	374	20 / 20	2,362,788	₹24,670.16 Cr
R-9 (Jalandhar)	248	220	18 / 20	900,746	₹20,183.04 Cr
R-8 (Jaipur)	266	255	20 / 20	1,321,549	₹19,899.02 Cr
R-4 (Chandimandir)	200	164	20 / 20	1,321,761	₹14,653.75 Cr
R-27 (Hisar)	209	199	20 / 20	1,007,338	₹13,817.05 Cr
R-29 (Delhi 2)	363	293	20 / 20	1,847,943	₹13,811.90 Cr
R-24 (Ambala)	216	189	19 / 20	627,355	₹8,405.26 Cr
R-30 (Bareilly)	161	152	20 / 20	876,537	₹6,609.09 Cr
R-3 (Pune)	223	198	19 / 20	644,364	₹4,624.67 Cr
R-6 (Lucknow)	119	103	20 / 20	259,235	₹4,524.97 Cr
R-7 (Kochi)	133	113	18 / 20	1,351,496	₹4,317.58 Cr
R-2 (Trivandrum)	116	101	20 / 20	1,854,159	₹4,068.13 Cr
R-5 (Secunderabad)	140	128	19 / 20	504,364	₹2,912.52 Cr
R-15 (Dehradun)	94	84	19 / 20	452,411	₹2,909.55 Cr
R-17 (Bangalore)	141	134	20 / 20	620,357	₹2,821.23 Cr

Table 3.2 — Top 15 States Ranked by Combined Cross-Pattern Exposure

State	Total Hospitals	Flagged Hospitals	Distinct Signals	Total Claims	Combined Exposure
Haryana	648	557	20 / 20	2,914,261	₹45,394.40 Cr
Punjab	595	489	19 / 20	1,762,883	₹30,008.92 Cr
Uttar Pradesh	660	558	20 / 20	2,626,689	₹24,613.52 Cr
Rajasthan	338	307	20 / 20	1,147,700	₹12,853.85 Cr
Delhi	588	425	20 / 20	1,609,427	₹10,637.46 Cr
Kerala	250	205	20 / 20	3,139,053	₹8,197.88 Cr
Chandigarh	85	54	18 / 20	775,171	₹7,035.26 Cr
Maharashtra	424	342	19 / 20	1,007,695	₹6,842.48 Cr
Uttarakhand	130	114	19 / 20	712,069	₹3,823.60 Cr

State	Total Hospitals	Flagged Hospitals	Distinct Signals	Total Claims	Combined Exposure
Andhra Pradesh	185	148	19 / 20	647,139	₹3,337.72 Cr
Tamil Nadu	195	164	19 / 20	556,906	₹3,146.07 Cr
Karnataka	175	151	20 / 20	623,190	₹2,833.31 Cr
West Bengal	106	80	18 / 20	247,579	₹2,467.16 Cr
Bihar	87	78	18 / 20	331,644	₹2,310.73 Cr
Nepal	36	29	18 / 20	285,322	₹1,663.62 Cr

Key Findings

- National cluster belt:** Haryana, Punjab, Uttar Pradesh, Rajasthan, Delhi — the top 5 states by combined cross-pattern exposure — together account for **71.5%** of all flagged exposure nationally, out of 35 states with any flagged hospitals. ECHS fraud is not spread evenly across India; it is heavily concentrated in this north/north-west belt.
- R-1 (Delhi)** has the largest combined cross-pattern exposure (₹24,670.16 Cr across 374 of its 429 hospitals flagged, on 20 of the 20 named signals) — corroborating the region-level deduction-rate finding in Pattern 1 with independent, claim-level evidence.
- Haryana** leads all states with ₹45,394.40 Cr in combined flagged exposure across 557 of its 648 hospitals — a geographic concentration that should anchor the next phase of regional field audits.

PATTERN 4

Multi-Unit Brand Concentration
Name-normalised hospital chain rollup

CROSS-PATTERN HOSPITAL CHAIN DETECTION

Description: Groups hospitals by normalised brand name (stripping legal-entity suffixes such as "(A Unit of ... Pvt Ltd)") to detect chains where multiple branches are independently flagged. **Total Units** is every empanelled hospital nationally matching that brand name; **Flagged Units** is the subset triggering at least one of the 20 named signals — a chain where most or all of its units are flagged is a stronger signature of deliberate, group-wide structuring than a chain with only one outlier branch. This analysis identified **564** multi-unit chains, of which the top 15 by combined exposure are shown below.

Table 4.1 — Top 15 Multi-Unit Chains by Combined Cross-Pattern Exposure

Chain	Total Units	Flagged Units	Distinct Signals	Combined Exposure
METRO HEART	10	9	18 / 20	₹3,732.26 Cr
PARK	5	5	16 / 20	₹2,668.60 Cr
MAX SUPER	17	15	17 / 20	₹2,472.76 Cr
LIVASA HOSP	2	2	14 / 20	₹2,264.20 Cr
PARK GURGAON	2	2	15 / 20	₹2,093.13 Cr
MEDANTA MEDICITY	2	2	15 / 20	₹2,082.85 Cr
HEALING TOUCH	4	4	15 / 20	₹1,961.46 Cr
SRI GURU	6	6	13 / 20	₹1,766.28 Cr
YATHARTH WELLNESS	4	4	16 / 20	₹1,582.70 Cr
EMC SUPER	2	2	14 / 20	₹1,439.91 Cr
PARAS HEALTHCARE	2	2	16 / 20	₹1,406.14 Cr
FORTIS MOHALI	4	2	12 / 20	₹1,312.75 Cr
IVY HEALTHCARE	2	2	13 / 20	₹1,262.16 Cr
STAR	2	2	15 / 20	₹1,151.89 Cr
PARK FARIDABAD	2	2	15 / 20	₹1,110.52 Cr

Key Findings

- METRO HEART** is the largest multi-unit cluster by combined exposure: **9 of its 10 units** nationally are independently flagged, across 18 of the 20 named signals — ₹3,732.26 Cr in combined flagged exposure.
- Chains where Flagged Units is close to Total Units (i.e. nearly every branch is flagged, not just one outlier) should be audited as a single corporate group rather than unit-by-unit — spreading activity across separately-registered legal entities under one brand makes per-hospital audit thresholds easier to stay under.

PATTERN 5

CROSS-REGION REFERRAL USAGE

Single-Hospital Funnelling
Referral-syndicate detection, Tukey-fenced

Description: Identifies polyclinics and referring entities that funnel most of their referrals to a single hospital. **Referrals** is the total count of referral transactions issued by that polyclinic. **Top Dest %** is the share of those referrals directed to its single most-favoured destination hospital (e.g. 80% means 8 in 10 referred patients all end up at the same hospital) — the higher this number, the more it looks like a steered pipeline rather than routine clinical choice. Each row is flagged **cross-region** when the polyclinic's city differs from the destination hospital's city. This analysis covers **78** funnelling pairs; **10** are statistically extreme by referral volume and **48** are cross-region.

Statistical threshold (Tukey fence): Applied to **referral count** across all 78 pairs — Q3 = 20,406 referrals, IQR = 18,296 referrals, fence = **47851.1250 referrals**. The **10 pairs** exceeding this threshold are statistically extreme in referral volume and are highlighted below.

Table 5.1 — Top 20 Funnelling Pairs by Exposure (Tukey-Fence Outliers Marked)

Polyclinic	City	Referrals	Top Dest %	Funnels To	Dest. City	Cross-Region	Exposure
Kochi	Kochi	238,623	64.6%	Amrita Institute of Medical Sc	Kochi	NO	₹292.86 Cr
VidyaDhar Nagar	VidyaDhar Nagar	89,748	67.5%	MANIPAL HEALTH ENTERPRISES PRI	JAIPUR	YES	₹262.94 Cr
Ludhiana	Ludhiana	144,948	57.5%	FORTIS HOSPITAL - LUDHIANA	LUDHIANA	NO	₹228.90 Cr
Belgaum	Belgaum	80,533	59.2%	KLES DR. PRABHAKAR KORE HOSPIT	BELAGAVI	YES	₹104.54 Cr
Palampur (RC Yol)	Palampur	112,007	68.1%	VIVEKANAND MEDICAL INSTITUTE	PALAMPUR	NO	₹95.87 Cr
Devlali	Devlali	66,756	63.7%	ASHOKA MEDICOVER HOSPITALS (UN	NASHIK	YES	₹76.45 Cr
Yelahanka	Bangalore	73,575	62.1%	ASTER CMI HOSPITAL (A UNIT OF	BANGALORE	NO	₹60.92 Cr
Kilimanur	Kilimanur	110,214	65%	SREE GOKULAM MEDICAL COLLEGE H	THIRUVANANTHAPURAM	YES	₹51.90 Cr
Vadodara	Vadodara	52,734	67.7%	NARHARI HOSPITAL	VADODARA	NO	₹51.33 Cr
Jagraon	Jagraon	37,702	56.2%	FORTIS HOSPITAL - LUDHIANA	LUDHIANA	YES	₹49.40 Cr
Mainpuri	Mainpuri	17,829	56.3%	PUSHPANJALI HOSPITAL & RESEARC	AGRA	YES	₹48.01 Cr
Udhampur	Udhampur	13,117	92.9%	NARAYANA VAISHNO DEVI SPECIALI	JAMMU & KASHMIR	YES	₹47.55 Cr
Fatehgarh	Fatehgarh	38,041	81%	REGENCY HOSPITAL LTD- KANPUR	KANPUR	YES	₹44.85 Cr
Sholapur	Sholapur	36,128	69.1%	Ashwini Sahakari Rugnalaya Ani	Solapur	YES	₹40.39 Cr

Polyclinic	City	Referrals	Top Dest %	Funnels To	Dest. City	Cross-Region	Exposure
COD Kandivali(Malad)	MUMBAI	48,678	59.6%	WOCKHARDT HOSPITALS LTD.	THANE	YES	₹33.85 Cr
Satna	Satna	33,618	78.1%	MP BIRLA HOSPITAL & PRIYAMVADA	SATNA	NO	₹27.08 Cr
Latur	Latur	39,428	72.9%	Vivekanand Medical Foundation	Latur	NO	₹26.01 Cr
Rajouri	Rajouri	10,669	55.4%	NARAYANA VAISHNO DEVI SPECIALI	JAMMU & KASHMIR	YES	₹25.96 Cr
Panaji (RC Pune)	Panaji	26,712	93%	MANIPAL HOSPITALS - GOA (A UNI	PANAJI	NO	₹22.75 Cr
Tirunelveli	Tirunelveli	34,230	68.8%	Shifa Hospitals	TIRUNELVELI	NO	₹21.79 Cr

Highlighted red = exceeds the Tukey-fence referral threshold of 47851.1250 referrals (statistical outlier)

Key Findings

- 10 funnelling pairs exceed the Tukey-fence referral threshold of 47,851 referrals, of which 5 are also cross-region — these are the highest-confidence referral-syndicate candidates in the dataset, combining statistical extremity with a geography that rules out routine convenience.
- 48 of the 78 funnelling pairs overall are cross-region (polyclinic and destination hospital in different cities). The largest cross-region pair is VidyaDhar Nagar (VidyaDhar Nagar) → MANIPAL HEALTH ENTERPRISES PRIVATE LIMITED (JAIPUR), worth ₹262.94 Cr.

CONSOLIDATED SUMMARY

All findings are based on structured database analysis and must be corroborated with physical audit records before enforcement action is taken.

#	Pattern	Coverage	Headline Finding
1	Regional Fraud Hotspots	31 regions	R-6 (Lucknow) — 19.23% deduction
2	Cross-Pattern Hospital Concentration	4,282 hospitals	SAKRA WORLD HOSPITAL — 18 of 20 named signals
3	Cross-Pattern Regional & State Clustering	31 regions, 35 states	Haryana — ₹45,394.40 Cr combined exposure
4	Cross-Pattern Hospital Chain Detection	564 multi-unit chains	METRO HEART — 9/10 units flagged
5	Cross-Region Referral Usage	78 funnelling pairs (10 Tukey outliers)	VidyaDhar Nagar → MANIPAL HEALTH ENTERPRISES PRIVATE LIMITED (67.5% of referrals)

Prepared by IIT Kanpur — Data Analytics & Fraud Intelligence Division | 1 July 2026
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